

Force Sensing for Continuum Robots: Analysis of Aloï's Method and Future Directions

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1 Executive Summary

This document analyzes the limitations of Aloï et al.'s Gaussian-based force estimation method for continuum robots and proposes a new approach that incorporates environmental knowledge to achieve unique and precise force estimation. The key insight is that while Aloï's method works well for certain scenarios, it fundamentally suffers from non-uniqueness issues when multiple solutions exist, 3D loads are present, or tangential friction forces occur.

2 Analysis of Aloï's Method (2022)

2.1 Core Methodology

Aloï's method estimates external loads on continuum robots using:

- **Input:** Fiber Bragg Grating (FBG) sensor measurements of robot shape/curvature
- **Parameterization:** External loads as sum of Gaussian distributions
- **Optimization:** Levenberg-Marquardt algorithm to minimize shape prediction error

The load distribution is modeled as:

$$f(s) = \sum_{i=1}^N A_i \exp\left(-\frac{(s - \mu_i)^2}{2\sigma_i^2}\right) \quad (1)$$

where $\theta = [A_1, \mu_1, \sigma_1, \dots, A_N, \mu_N, \sigma_N]^T$ are the parameters to be estimated.

2.2 Optimization Formulation

The method solves:

$$\min_{\theta} J(\theta) = \|\mathbf{y}_{\text{meas}} - \mathbf{y}_{\text{pred}}(\theta)\|^2 + \lambda \|\theta - \theta_0\|^2 \quad (2)$$

where:

- $\mathbf{y}_{\text{meas}}$: measured robot shape from FBG sensors
- $\mathbf{y}_{\text{pred}}(\theta)$: predicted shape from Cosserat rod model with load $f(s)$
- λ : regularization weight
- θ_0 : prior estimate

2.3 Strengths

1. **Smooth representation:** Gaussian parameterization provides continuous, differentiable load distribution
2. **Computational efficiency:** Low-dimensional parameter space (6 parameters for 2 loads)
3. **Good shape reconstruction:** Achieves excellent shape RMSE (< 0.15 mm in our tests)
4. **Handles distributed loads:** Natural representation for contact forces

2.4 Fundamental Limitations

2.4.1 1. Non-Uniqueness of Solutions

Problem: Multiple load configurations can produce identical robot shapes.

Mathematical Basis: The inverse problem is ill-posed. Given shape measurements $\mathbf{y}$, the mapping $f(s) \rightarrow \mathbf{y}$ is not injective. Different load distributions can satisfy:

$$\mathbf{y}_{\text{pred}}(f_1) = \mathbf{y}_{\text{pred}}(f_2) \quad \text{but} \quad f_1 \neq f_2 \quad (3)$$

Example Scenarios:

- Two small loads vs. one large load at intermediate position
- Distributed load vs. concentrated load with same moment
- Loads at different positions with compensating magnitudes

Evidence from Our Implementation:

- Case 1: Load 2 magnitude error 91.7% (before multi-start optimization)
- Case 3: Load 1 position error 40mm (13% of robot length)
- Multi-start optimization (3600 initial guesses) required to find good solutions

2.4.2 2. Point Load vs. Gaussian Mismatch

Problem: Real contact forces are often point loads, but Gaussian representation is inherently distributed.

Consequence:

- Cannot precisely localize point contacts
- Estimated load “spreads out” over $\pm 3\sigma$ region
- Position estimates are centroids, not actual contact points

Our Results: Even with optimal parameters, estimated loads show:

- $\sigma \approx 10 - 20$ mm (3-7% of robot length)
- Position errors up to 15mm for small loads
- Magnitude errors when loads are close together (Gaussian overlap)

2.4.3 3. 2D Planar Assumption

Problem: Method assumes all loads are normal to robot backbone in a single plane.

Limitations:

- Cannot distinguish 3D out-of-plane loads
- Cannot estimate tangential (friction) forces along backbone
- Cannot handle torsional loads

Real-World Impact: Most manipulation tasks involve:

- 3D contact forces (e.g., grasping, pushing objects)
- Friction during sliding contact
- Combined normal + tangential loads

2.4.4 4. Local Minima in Optimization

Problem: Non-convex objective function with many local minima.

Manifestation:

- Sensitive to initial guess θ_0
- Can converge to wrong load configuration
- Requires multi-start strategy (computationally expensive)

Our Solution: Implemented 3600 initial guesses ($4 \text{ forces} \times 5 \text{ positions} \times 3 \text{ widths} \times 2 \text{ loads}$), but this:

- Increases computation time $3600\times$
- Still no guarantee of global optimum
- Not practical for real-time applications

2.4.5 5. Number of Loads Must Be Known A Priori

Problem: Must specify N (number of Gaussians) before optimization.

Consequence:

- If N too small: cannot represent actual load distribution
- If N too large: overfitting, spurious loads appear
- No automatic model selection mechanism

3 Representative Failure Scenarios

3.1 Scenario 1: Multiple Solutions Exist

Setup: Robot contacts a flat surface at unknown location.

Problem:

- Single normal force at position s_1 produces shape $\mathbf{y}$
- But distributed load over $[s_1 - \Delta, s_1 + \Delta]$ can produce similar $\mathbf{y}$
- Aloï's method may converge to either solution

Why It Fails: No mechanism to prefer point contact over distributed contact.

3.2 Scenario 2: 3D Loads

Setup: Robot pushes against object at angle (force has x, y, z components).

Problem:

- FBG sensors measure bending in two planes: $\kappa_x(s), \kappa_y(s)$
- But cannot measure torsion $\tau(s)$ or axial strain $\epsilon(s)$
- 3D force $\mathbf{f} = [f_x, f_y, f_z]^T$ projects to 2D measurements

Why It Fails: Underdetermined system - 3 unknowns, 2 measurements per point.

3.3 Scenario 3: Tangential Friction

Setup: Robot slides along surface with friction.

Problem:

- Normal force $f_n(s)$ causes bending
- Tangential force $f_t(s)$ causes compression/tension
- FBG measures only curvature, not axial strain

Why It Fails: Tangential forces are invisible to curvature-based estimation.

3.4 Scenario 4: Close Multiple Contacts

Setup: Robot grasps object with two contact points 2cm apart.

Problem:

- Two Gaussians with $\sigma \approx 1.5$ cm overlap significantly
- Optimization struggles to separate them
- Often converges to single larger Gaussian at midpoint

Why It Fails: Gaussian basis functions are not orthogonal; interference between nearby loads.

4 Proposed Solution: Environment-Informed Force Estimation

4.1 Core Idea

Fundamental Insight: Incorporate knowledge of the environment to constrain the solution space and achieve unique, precise force estimation.

$$\boxed{\text{FBG Measurements} + \text{Environment Knowledge} \rightarrow \text{Unique Force Estimate}} \quad (4)$$

4.2 Types of Environmental Knowledge

4.2.1 1. Geometric Constraints

Information: Known object shapes, surface locations, workspace boundaries.

Application:

- Contact can only occur at robot-object intersection points
- Force direction must be normal to contact surface
- Reduces search space from continuous $s \in [0, L]$ to discrete contact points

Mathematical Formulation:

$$s_i \in \mathcal{S}_{\text{contact}} = \{s : \|\mathbf{r}(s) - \mathbf{r}_{\text{obj}}(s)\| < \epsilon\} \quad (5)$$

4.2.2 2. Force Direction Constraints

Information: Contact normal vectors from object geometry.

Application:

- Force must be along surface normal: $\mathbf{f}_i = f_i \mathbf{n}_i$
- Reduces 3D force to 1D magnitude estimation
- Eliminates tangential component ambiguity

Mathematical Formulation:

$$\mathbf{f}(s_i) = f_i \mathbf{n}_{\text{obj}}(\mathbf{r}(s_i)) \quad \text{where} \quad f_i \geq 0 \quad (6)$$

4.2.3 3. Contact Model Constraints

Information: Physics-based contact models (Hertz, Hunt-Crossley, etc.).

Application:

- Relate force magnitude to penetration depth: $f = k\delta^n$
- Constrain force-displacement relationship
- Provide prior on expected force magnitudes

4.2.4 4. Friction Cone Constraints

Information: Coulomb friction coefficient μ .

Application:

- Tangential force bounded: $|f_t| \leq \mu f_n$
- Enables estimation of both normal and tangential components
- Prevents physically impossible force configurations

Mathematical Formulation:

$$\|\mathbf{f}_t\| \leq \mu \|\mathbf{f}_n\| \quad \Rightarrow \quad \mathbf{f} \in \text{Friction Cone} \quad (7)$$

4.3 Proposed Method Formulation

4.3.1 Problem Statement

Given:

- FBG measurements: $\mathbf{y}_{\text{meas}} = [\kappa_x(s_1), \kappa_y(s_1), \dots, \kappa_x(s_M), \kappa_y(s_M)]^T$
- Environment model: object geometry $\mathcal{O}$, contact model $\mathcal{C}$
- Robot model: Cosserat rod with parameters (L, EI, GJ, EA)

Estimate:

- Number of contacts: N
- Contact locations: $\mathbf{s} = [s_1, \dots, s_N]^T$
- Contact forces: $\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_N]^T$ (3D vectors)

4.3.2 Optimization Formulation

$$\begin{aligned} \min_{N, \mathbf{s}, \mathbf{F}} \quad J = & \underbrace{\|\mathbf{y}_{\text{meas}} - \mathbf{y}_{\text{pred}}(\mathbf{s}, \mathbf{F})\|^2}_{\text{Shape fitting term}} + \underbrace{\lambda_1 \sum_{i=1}^N \|\mathbf{f}_i\|^2}_{\text{Force regularization}} \\ \text{subject to} \quad & s_i \in \mathcal{S}_{\text{contact}}(\mathcal{O}) \quad \forall i \\ & \mathbf{f}_i \cdot \mathbf{n}_i \geq 0 \quad (\text{no tension}) \\ & \|\mathbf{f}_{i,t}\| \leq \mu \|\mathbf{f}_{i,n}\| \quad (\text{friction cone}) \\ & f_i = \mathcal{C}(\delta_i) \quad (\text{contact model}) \\ & \delta_i = \text{penetration depth at } s_i \end{aligned} \quad (8)$$

4.3.3 Key Advantages Over Aloï's Method

1. **Uniqueness:** Geometric constraints eliminate spurious solutions
2. **Precision:** Point contact model (not Gaussian) for accurate localization
3. **3D Capability:** Estimates full 3D force vectors
4. **Friction Handling:** Explicitly models tangential forces
5. **Automatic N Selection:** Contact detection from geometry

4.4 Implementation Strategy

4.4.1 Phase 1: Contact Detection

1. Predict robot shape from FBG: $\mathbf{r}(s) = \int_0^s \mathbf{R}(s') \mathbf{e}_3 ds'$
2. Find intersections with environment: $\mathcal{S}_{\text{contact}} = \{s : d(\mathbf{r}(s), \mathcal{O}) < \epsilon\}$
3. Initialize $N = |\mathcal{S}_{\text{contact}}|$ contact points

4.4.2 Phase 2: Force Estimation

For each contact point s_i :

1. Compute surface normal: $\mathbf{n}_i = \nabla d(\mathbf{r}(s_i), \mathcal{O})$
2. Compute penetration: $\delta_i = -d(\mathbf{r}(s_i), \mathcal{O})$
3. Initialize force: $f_{i,n}^{(0)} = k\delta_i^{3/2}$ (Hertz model)
4. Optimize: $\min_{\mathbf{f}_i} \|\mathbf{y}_{\text{meas}} - \mathbf{y}_{\text{pred}}\|^2$ subject to friction cone

4.4.3 Phase 3: Refinement

1. Jointly optimize all contact locations and forces
2. Use gradient-based method (L-BFGS-B with box constraints)
3. Validate solution: check force balance, moment balance

4.5 Required Inputs

Input	Source	Precision Required
Object geometry	CAD model / 3D scan	± 1 mm
Robot base pose	Robot kinematics	± 2 mm, $\pm 2^\circ$
FBG measurements	Fiber sensors	± 0.1 mm (shape)
Material properties	Calibration	$\pm 10\%$ (EI, GJ)
Friction coefficient	Lookup table	± 0.1

Table 1: Required environmental knowledge and precision

5 Validation Plan

5.1 Simulation Experiments

1. **Single contact:** Flat surface, varying angles
2. **Multiple contacts:** Two-point grasp, three-point support
3. **3D loads:** Out-of-plane forces, combined loading
4. **Friction:** Sliding contact, tangential forces
5. **Complex geometry:** Curved surfaces, corners, edges

5.2 Comparison Metrics

- **Position error:** $e_s = |s_{\text{est}} - s_{\text{true}}|$
- **Magnitude error:** $e_f = ||\mathbf{f}_{\text{est}}|| - ||\mathbf{f}_{\text{true}}||$
- **Direction error:** $e_\theta = \arccos(\hat{\mathbf{f}}_{\text{est}} \cdot \hat{\mathbf{f}}_{\text{true}})$
- **Shape RMSE:** $\text{RMSE} = \sqrt{\frac{1}{M} \sum_{j=1}^M \|\mathbf{r}_{\text{est}}(s_j) - \mathbf{r}_{\text{true}}(s_j)\|^2}$

5.3 Baseline Comparisons

- Aloï’s Gaussian method (current implementation)
- Rucker’s EKF method
- Pure FBG shape sensing (no force estimation)

6 Expected Outcomes

6.1 Quantitative Improvements

Metric	Aloi Method	Proposed Method
Position error	10-40 mm	< 5 mm
Magnitude error	10-90%	< 15%
3D force estimation	No	Yes
Friction estimation	No	Yes
Uniqueness guarantee	No	Yes (with env.)

Table 2: Expected performance comparison

6.2 Qualitative Advantages

- **Physical interpretability:** Forces correspond to actual contacts
- **Robustness:** Fewer local minima due to constrained search space
- **Scalability:** Handles arbitrary number of contacts automatically
- **Real-time capability:** Faster convergence with good initialization

7 Limitations and Future Work

7.1 Limitations of Proposed Method

1. **Requires environment model:** Not applicable to unknown environments
2. **Sensitive to model errors:** Inaccurate geometry $\rightarrow$ wrong contact locations
3. **Assumes rigid objects:** Deformable objects need different contact model
4. **Computational cost:** Contact detection + optimization may be slow

7.2 Future Extensions

1. **Hybrid approach:** Use Aloï for unknown environments, proposed method when environment known
2. **Online learning:** Update environment model from repeated contacts
3. **Uncertainty quantification:** Bayesian framework for force estimation
4. **Dynamic forces:** Extend to time-varying loads, impact forces

8 Conclusion

Aloï’s Gaussian-based method provides a strong foundation for force estimation but suffers from fundamental limitations:

- Non-unique solutions
- Point load vs. Gaussian mismatch
- 2D planar assumption
- Cannot handle friction

The proposed environment-informed approach addresses these limitations by:

- Constraining solution space with geometric knowledge
- Using point contact model for precision
- Estimating full 3D forces including friction
- Providing uniqueness guarantees

Next Steps for Meeting:

1. Discuss which failure scenarios are most critical for target applications
2. Determine availability of environment models (CAD, sensors, etc.)
3. Define validation experiments and success criteria
4. Assign implementation tasks and timeline